

NMAM INSTITUTE OF TECHNOLOGY, NITTE
 (An Autonomous Institution affiliated to VTU, Belagavi)
Sixth Semester B.E. (E&C) (Credit System) Degree Examinations
 May - June 2023

Duration: 3 Hours

20ECE55 - MACHINE LEARNING AND ITS APPLICATIONS

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1. a)	With a neat block diagram, discuss different component of a pattern recognition system.	10	L*2	1	1
b)	Discuss on the process of training and testing in machine learning with a suitable example.	10	L2	1	1
2. a)	Explain reinforcement learning with a neat block diagram.	10	L2	1	1
b)	Discuss the concept of principal component analysis with an example.	10	L3	1	1
3. a)	Explain the concept of dimensionality reduction. State any two dimensionality reduction methods with an example each.	10	L3	1	1
b)	Discuss in brief with block diagrams and equations how Independent component analysis can be used in dimensionality reduction.	10	L3	1	1
Unit – II					
4. a)	Discuss the application of multi-variate data analysis in machine learning.	10	L2	3	1
b)	Differentiate between two-class classification and multiclass classification problem.	10	L3	3	1
5. a)	Explain k-fold cross validation with respect to classification.	10	L3	3	1
b)	Discuss confusion matrix for a two class classification problem. Discuss its elements. How accuracy can be computed from confusion matrix?	10	L2	4	1
6. a)	State any two statistical data visualization tools and explain them in detail.	8	L2	4	1
b)	Define sensitivity, specificity and accuracy for a two class classification problem. Compute them for following confusion matrix.				
	$\begin{bmatrix} 96 & 4 \\ 3 & 98 \end{bmatrix}$	12	L3	4	1
Unit – III					
7. a)	Discuss in detail the decision tree classifier.	10	L3	5	1
b)	Discuss the user interface and native regression tools in WEKA software for machine learning.	10	L3	5	1
8. a)	Explain with necessary expressions the SVM classification problem.	10	L3	5	1
b)	Discuss the working principle of k-nearest neighbour classifier model.	10	L3	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
